



Research article

Calcium regulates the cell-to-cell water flow pathway in maize roots during variable water conditions

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ABSTRACT

Soil water shortages can decrease root hydraulic conductivity and affect Ca uptake and movement through the plant. In this study, the effects of extra Ca^{2+} applied in nutrient solution on the hydraulic properties of the whole roots (L_{pr}) and cortical cells ($L_{p_{cell}}$) of maize (*Zea mays* L.) subjected to variable water conditions were investigated. Under well-watered conditions, extra Ca^{2+} significantly increased the root Ca content, total root length, and lateral root number; however, it reduced the root cortical cell volume, L_{pr} , and $L_{p_{cell}}$. Hg^{2+} inhibition experiments suggested that extra Ca^{2+} could reduce the contribution of the cell-to-cell water flow pathway. Osmotic stress (10% PEG6000) significantly decreased the cortical cell volume, L_{pr} , and $L_{p_{cell}}$ in the control plants, but smaller decreases were observed in the extra Ca^{2+} plants. The Hg^{2+} treatment reduced the L_{pr} larger in the extra Ca^{2+} plants (74.6%) than in the control plants (53.2%), suggesting a higher contribution of the cell-to-cell pathway. The larger Hg^{2+} inhibition of the $L_{p_{cell}}$ in the extra Ca^{2+} roots (67.2%) when compared to the controls (56.4%) indicated that extra Ca^{2+} can mitigate the inhibition of aquaporin expression and/or activity levels via osmotic stress. After 2 d of rehydration, the extra Ca^{2+} helped the L_{pr} and $L_{p_{cell}}$ to recover almost completely, but these properties only partially recovered in the control plants. In conclusion, extra Ca^{2+} may adjust the contribution of cell-to-cell pathway by regulating the expression and/or activity levels of AQPs according to water availability; this regulation may weaken negative effects and optimize water use.

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1. Introduction

Calcium (Ca) is an essential plant macronutrient that plays key roles in cell wall and membrane structure. Calcium ions (Ca^{2+}) act as secondary messengers for many signals [1–3]. Water and solutes can be transported within plants through apoplastic, symplastic and transcellular pathways. In many plants, Ca enters the root apoplast with the mass flow of water as a divalent cation (Ca^{2+}) from the soil [4]. In addition, Ca can enter root cells through a variety of Ca^{2+} -permeable cation channels [5]. However, Ca^{2+} apparently travels apoplastically across the root cortex before it

reaches the Casparian band; this route is necessary to enter the xylem via endodermal cell cytoplasm [6]. The non-xylem apoplast compartment may significantly affect ion concentrations within tissues especially because the diffusion rates of solutes and water are different [7]. Plants transport Ca through the xylem to the shoot as either Ca^{2+} or complexed with other organic acids, and the cytosolic Ca concentration ($[\text{Ca}^{2+}]_{\text{cyt}}$) is sensitive to the apoplastic Ca concentration ($[\text{Ca}^{2+}]_{\text{apo}}$) [5]. Moreover, the symplastic pathway of Ca delivery allows plants to control Ca transport rate and selectivity, being essential for cell signaling [3]. Therefore, transpiration-regulated water flow may correlate with Ca delivery and storage both at the organ and cell water relation levels, and aquaporins (AQPs) may be involved [2].

The extracellular matrix (ECM) is the major component of the cell wall apoplast, and the $[\text{Ca}^{2+}]_{\text{apo}}$ helps to strengthen the ECM [8]. However, the ECM structure can affect the apoplastic pathway of water and Ca^{2+} transport [2]. When leaf transpiration is high, the apoplastic water flow dominates the whole water flow, and Ca delivery through the apoplast can be enhanced [9]. When the intercellular conditions are balanced, the $[\text{Ca}^{2+}]_{\text{cyt}}$ is finely controlled at approximately 100 nM [10]. Because $[\text{Ca}^{2+}]_{\text{cyt}}$ is sensitive to many developmental cues and/or environmental

Abbreviations: AQP, Aquaporin; Ca, Calcium; Ca^{2+} , Calcium ion; $[\text{Ca}^{2+}]_{\text{apo}}$, Apoplastic calcium concentration; $[\text{Ca}^{2+}]_{\text{cyt}}$, Cytosolic calcium concentration; *E*, Leaf transpiration rate; *g_s*, Leaf stomatal conductance; HgCl_2 , Mercuric chloride; L_{pr} , Whole root hydraulic conductivity; $L_{p_{cell}}$, Cell hydraulic conductivity.

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stimuli, it is considered to be essential for the production of physiological responses [3]. Furthermore, it is thought that the $[Ca^{2+}]_{cyt}$ regulates water flow through aquaporins (AQPs) [11]. The $[Ca^{2+}]_{apo}$ can affect stomatal aperture, and AQPs in the guard cells may be involved in this process [12].

Total root hydraulic conductivity includes vascular (the xylem in the stele) and non-vascular components (before water enters the stele). AQPs may be involved in these two components during water transport across the cortex or parenchyma cells adjacent to the xylem [13]. AQPs belong to the family of major intrinsic proteins (MIPs) and can finely regulate water flow across membranes [14]. Studies have indicated that AQP activity levels can be regulated by phosphorylation following $[Ca^{2+}]_{cyt}$ changes [11,15]. When purified AtPIP2; 1 was reconstituted in proteoliposomes, Ca^{2+} could bind and/or gate AtPIP2; 1, inhibiting water permeability [16]. This result suggests that Ca^{2+} directly affects AQPs; however, Ca^{2+} -dependent protein kinases and phosphatases are also involved [2]. Another study also demonstrated that the addition of Ca^{2+} could down regulate the hydraulic conductivity of *Arabidopsis* plasma membrane vesicles [17]. However, Ca^{2+} has been reported to ameliorate the decrease in the roots and cell hydraulic conductivity under salt stress [18], and this was thought to be related to AQP function [19,20]. A recent study demonstrated that increasing the $[Ca^{2+}]_{apo}$ by reducing the capacity for mesophyll Ca^{2+} accumulation could enhance cell wall thickness and lower the extensibility, transpiration and growth rate of the cell wall [21]. This result suggests that hydraulic responses to transient levels of extra Ca^{2+} may be due to an increase in the rapidly changeable $[Ca^{2+}]_{cyt}$, but a long-term high $[Ca^{2+}]_{apo}$ may alter the cell wall structure and inhibit cell-to-cell water flow.

Because the interactions among Ca, water flow pathways, and AQP function are unclear, further studies are required [2]. Water stress can reduce the hydraulic conductivity of roots [22], but it is unclear whether Ca can compensate this negative effect. This report describes the outcome of studies regarding the effects of calcium treatments on the regulation of water transport through maize roots that were grown under water stress and rehydration conditions. The effect of mercury chloride ($HgCl_2$) on cortex cell osmotic water permeability was also measured using a cell pressure probe.

2. Results

2.1. Root calcium contents

The root apex (0–1 cm) contained the largest calcium content among the root zones; water stress significantly increased the Ca content of the 0–1 cm root segments (Fig. 1). The extra Ca^{2+} increased the Ca content across portions of the entire root.

2.2. Leaf gas exchange

Leaf stomatal conductance (g_s) responded rapidly to changing water conditions. Osmotic stress significantly decreased g_s , and 2 d of rehydration led to a partial recovery. Extra Ca^{2+} did not significantly affect g_s when the plants were well-watered, and the g_s of the extra Ca^{2+} plants did not change significantly during osmotic stress or rehydration (Fig. 2A). Leaf transpiration (E) of the control Ca^{2+} plants showed similar changes with g_s during osmotic stress and rehydration, but it completely recovered. For the extra Ca^{2+} plants, E was stable across the different water conditions and was significantly higher than that of the controls (Fig. 2B).

2.3. Hydraulic conductivity of the entire root (L_{pr})

The L_{pr} of the well-watered plants that were supplied with extra Ca^{2+} (11 mM) was significantly lower ($P < 0.05$) than that of the

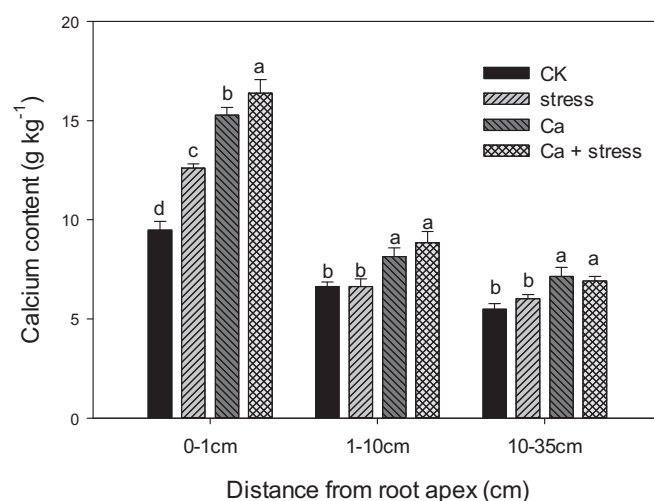


Fig. 1. Effects of extra calcium in nutrient solution and water stress on the Ca content of maize roots. CK = 1 mM Ca^{2+} in nutrient solution; stress = 10% PEG-induced water stress in 1 mM Ca^{2+} nutrient solution; Ca = 11 mM Ca^{2+} ; Ca + stress = 10% PEG-induced water stress in 11 mM Ca^{2+} nutrient solution. The data are the means $\pm$ SE. The different lowercase letters above the bars denote significant differences ($P < 0.05$).

controls (Fig. 3A). Water stress significantly reduced L_{pr} by 53.79% ($P < 0.05$) in the controls. However, extra Ca^{2+} lessened this reduction. No significant difference in the L_{pr} of the high Ca^{2+} plants was observed regardless of the water treatment (well-watered, water stress or subsequent rehydration); the L_{pr} was approximately $4.5 \times 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$ (Fig. 3ABC).

For the well-watered plants, $HgCl_2$ (50 μM) caused sharp declines in L_{pr} in the extra Ca^{2+} level plants (62.1%); this result was similar to the control Ca^{2+} treatment (62.2%). If the Hg^{2+} -induced decline can be considered as an indicator to the cell-to-cell water flow pathway, our data suggest that the cell-to-cell pathways may be inhibited by extra Ca^{2+} in well-watered plants (Fig. 3A). This decline was reversed by a subsequent treatment with 5 mM β -mercaptoethanol (Fig. 3A). Mercaptoethanol did not significantly affect the L_{pr} of roots (Supplementary Fig. A) that did not receive the Hg^{2+} treatment. Osmotically stressed, Hg^{2+} treatment caused a larger L_{pr} reduction (74.6%) in the extra Ca^{2+} plants than in the control plants (Fig. 3B). The L_{pr} of the stressed plants subjected to the Hg^{2+} treatment was approximately $1 \times 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$; this value was significantly lower than that of the well-watered plants (approximately $2 \times 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$), which might suggest osmotic stress reduced the apoplastic pathway and/or the water permeability of Hg^{2+} insensitive aquaporins and lipid membrane. These pathways were also inhibited by calcium, since L_{pr} was further reduced by Ca^{2+} in osmotically stressed and Hg -treated plants (Fig. 3AB). Nevertheless, the overall results suggest that osmotic stress reduces cell-to-cell water flow and extra Ca^{2+} may enhance the involvement of the cell-to-cell pathway during osmotic stress. After 2 d of rehydration, Hg^{2+} displayed a higher L_{pr} inhibition in the plants that were treated with extra Ca^{2+} (62.4%) when compared to the controls (55.2%), suggesting a higher cell-to-cell water flow pathway (Fig. 3C).

2.4. Root morphology

Extra Ca^{2+} significantly increased the total root length and number of lateral roots in the well-watered plants (Fig. 4AD). However, there was no significant difference in the total root surface area of the plants that received the two different Ca^{2+} treatments (Fig. 4B). Although water stress significantly decreased

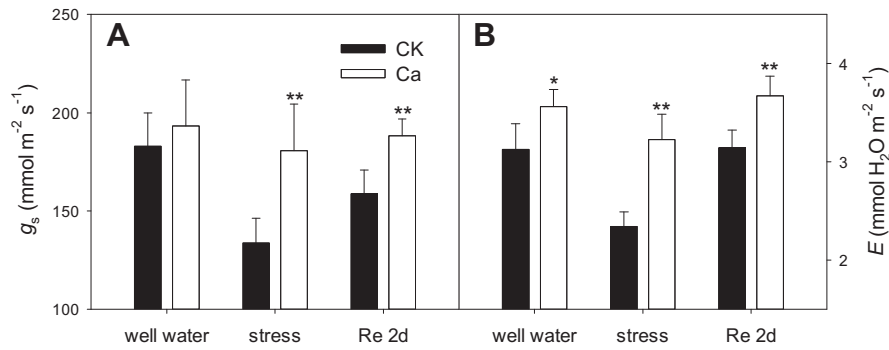


Fig. 2. Effects of extra calcium in nutrient solution on the leaf stomatal conductance (A) and transpiration (B) of the third leaf. CK (black box) = 1 mM Ca²⁺ in nutrient solution and Ca (white box) = 11 mM Ca²⁺. Stress = 10% PEG-induced water stress and Re 2d = rehydration for 2 days. A, total root length; B, total root surface area; C, primary root diameter; D, lateral root number. The data are the means of 5 or 6 plants with SE. Levels of significance: * $P \leq 0.05$; ** $P < 0.01$.

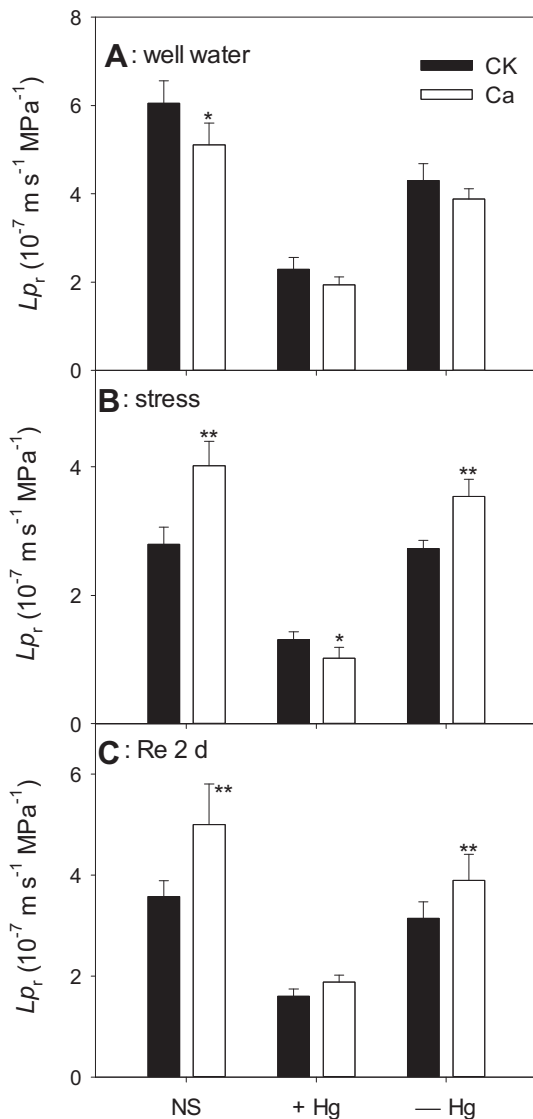


Fig. 3. Inhibition of whole root hydraulic conductivity (L_{pr}) of maize seedlings by Hg²⁺. A, well-watered conditions; B, 10% PEG-6000-induced water stress; C, rehydration for 2 days. CK (black box) = 1 mM Ca²⁺ in nutrient solution and Ca (white box) = 11 mM Ca²⁺. NS = nutrient solution; +Hg = nutrient solution with 50 μM Hg²⁺; -Hg = Hg²⁺ was removed from the nutrient solution containing 5 mM β -mercaptoethanol. The data are the means $\pm$ SE and $N = 7$ or 8. Levels of significance: * $P \leq 0.05$; ** $P < 0.01$.

the total root length, root surface area and number of lateral roots, extra Ca²⁺ significantly reduced this decline (Fig. 4ABD). Moreover, extra Ca²⁺ significantly increased the primary root diameter, which was unaffected by water stress (Fig. 4C). Within 2 d of rehydration, these four morphological parameters did not differ significantly between the rewatered and stressed plants. Therefore, the higher calcium level mitigated the negative effect of water stress on root morphology, and 2 d of rehydration was too short for significant morphological changes to occur.

2.5. Cortex cell dimensions

The diameter of the cortex cells was the highest in the 4–6 and 7–9 layers. In the well-watered plants, extra Ca²⁺ did not affect the cortex cell diameter. Although osmotic stress reduced the 7–9 layers cell diameters of the control plants, extra Ca²⁺ inhibited this reduction (Fig. 5A). However, extra Ca²⁺ did not affect cortex cell length significantly for either the well-watered or stressed plants (Fig. 5B). The cell volumes were calculated from the means of the cell diameters and lengths. Interestingly, osmotic stress decreased the cortex cell volume of all of the cell layers in the control plants. The addition of 10 mM Ca²⁺ also decreased the cell volumes in the well-watered plants, but it significantly increased the volumes during water stress (Fig. 5C). The cell volume changes may be related to cell-to-cell water flow, which was enhanced by the addition of extra Ca²⁺ during water stress (Fig. 3B).

2.6. Hydraulic properties of cortical cells

The turgor of the primary root cortical cells differed little across the layers (data not shown). Therefore, only the data from cortical cell layers 4–6 are shown in Fig. 6. Cell turgor declined significantly during osmotic stress ($P < 0.01$) and recovered following rehydration, but no responses to extra Ca²⁺ were observed (Fig. 6A). Extra Ca²⁺ decreased the elastic modulus of the cortex cell wall in the well-watered and stressed plants, but it increased the elastic modulus following rehydration when compared with the controls. Osmotic stress significantly increased the elastic modulus, and the elastic modulus declined to the well-watered levels after 2 d of rehydration ($P < 0.01$) (Fig. 6B).

To further confirm the hypothesis that extra Ca²⁺ could regulate cell-to-cell pathway (Fig. 3), the hydraulic conductivity of cortex cells ($L_{p_{cell}}$) was measured with a cell pressure probe. The $L_{p_{cell}}$ values were similar with the L_{pr} (Fig. 3; Fig. 7). For the well-watered plants, extra Ca²⁺ significantly reduced the $L_{p_{cell}}$ of the cortex cells, and Hg²⁺ induced a smaller decrease (62.6%) in the $L_{p_{cell}}$ of the extra Ca²⁺ plants when compared to the controls (72.9%). These

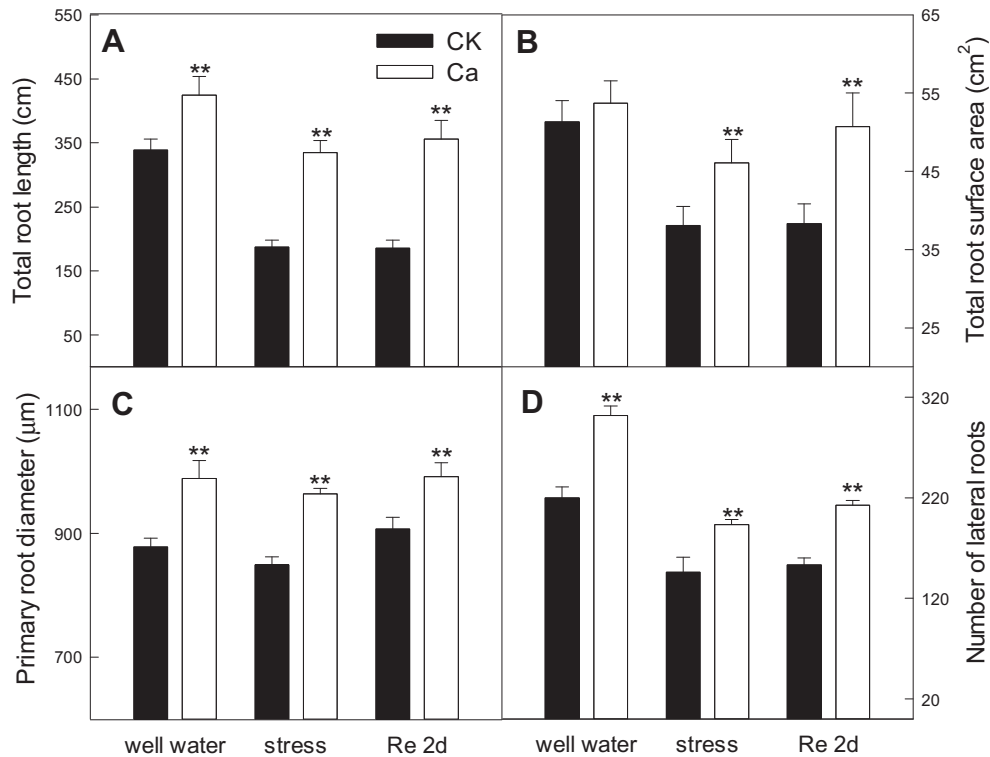


Fig. 4. Effects of extra calcium in nutrient solution on maize root morphology. CK (black box) = 1 mM Ca^{2+} in nutrient solution and Ca (white box) = 11 mM Ca^{2+} . Stress = 10% PEG-induced water stress and Re 2d = rehydration for 2 days. A, total root length; B, total root surface area; C, primary root diameter; D, lateral root number. The data are the means $\pm$ SE and $N = 5$. Level of significance: ** $P < 0.01$.

data suggest that reduced aquaporin quantities or activities may play a key role in reducing the $L_{p_{\text{cell}}}$ in the presence of extra Ca^{2+} (Fig. 7A). Osmotic stress reduced the $L_{p_{\text{cell}}}$, but extra Ca^{2+} mitigated this reduction by likely regulating AQPs. Hg^{2+} inhibited $L_{p_{\text{cell}}}$ to a larger extent in the extra Ca^{2+} roots (67.2%) than in the controls (56.4%). The $L_{p_{\text{cell}}}$ in the extra Ca^{2+} roots was significantly higher than the control plants only in the absence of Hg^{2+} (Fig. 7B). After 2 d of rehydration, extra Ca^{2+} helped the $L_{p_{\text{cell}}}$ recover to the levels of the well-watered plants (93.6%); only a 59.1% recovery was observed in the controls. Hg^{2+} treatments also revealed that extra Ca^{2+} induced enhanced membrane permeability, which was controlled by AQPs (Fig. 7C). These results suggest that extra Ca^{2+} counteracted the inhibition of AQP activity and/or the decrease of the quantity of AQPs to enhance the relative contribution of the cell-to-cell water pathway during osmotic stress and rehydration; opposing effects were observed in the well-watered conditions.

3. Discussion

Drought stress confers large ecological and physiological impacts on plant growth. Water stress may affect competitive Ca uptake and transport or partitioning within the plant [2]. Calcium plays an essential role in the maintenance and regulation of various cell functions, and the availability of Ca^{2+} to cells seems to be of fundamental importance in determining a plant's water stress tolerance [23,24]. The present study was conducted to determine how calcium regulates hydraulic conductivity under osmotic stress and rehydration conditions. The results indicated that extra Ca^{2+} can reduce water loss during well-watered conditions, and it can abate the decline in the hydraulic conductivity of water-stressed roots by enhancing the contribution of the cell-to-cell water flow pathway.

The presence of extra Ca in the nutrient solution increased the Ca content in the roots (especially in the root apex) (Fig. 1); this result may have occurred because mineral ion uptake primarily occurs at the root tips [25]. Studies have demonstrated that salt stress can induce a Ca deficit due to the competitive uptake of sodium, transport or partitioning [26,27]. However, our data suggested that osmotic stress increased the Ca content in the roots (Fig. 1). Osmotic stress reduced the hydraulic conductivity of the entire root and cortex cells (Fig. 3; Fig. 7); reduced water flow may inhibit Ca movement and help the plants to accumulate more Ca in the root tips (Fig. 1).

In the presence of extra Ca^{2+} , significant decreases in L_{p_r} and $L_{p_{\text{cell}}}$ and increased transpiration rates were observed in the well-watered plants. These conflicting data may be because that only the transient and partial leaf transpiration was measured, but not the transpiration of whole plants. Because the leaf area was reduced due to the higher extracellular Ca^{2+} concentration [21], partial leaf transpiration may be overestimated and may be inconsistent with the whole plant transpiration rate. Water flow is known to be primarily apoplastic in hydrostatic experiments; the flow bypasses the root protoplasts [28]. Higher extracellular Ca^{2+} levels may induce a thicker cell wall and smaller cell size [21], which may help to increase the contribution of apoplastic pathway. The direct application of Ca^{2+} to media containing purified *Arabidopsis* and *Beta vulgaris* membrane vesicles decreased the osmotic water permeability coefficient in the vesicles [11,17]. Furthermore, a structure-function analysis of AtPIP2; 2 and AtPIP2; 1 in *Arabidopsis* indicated that Ca^{2+} could directly close these two water channels [16,29]. Thus, as shown in this study, extra Ca^{2+} may increase the contribution of the apoplastic pathway in well-watered plants. Mercury ions (Hg^{2+}) can inhibit water channel flow and may be used to evaluate the contribution of the cell-to-cell

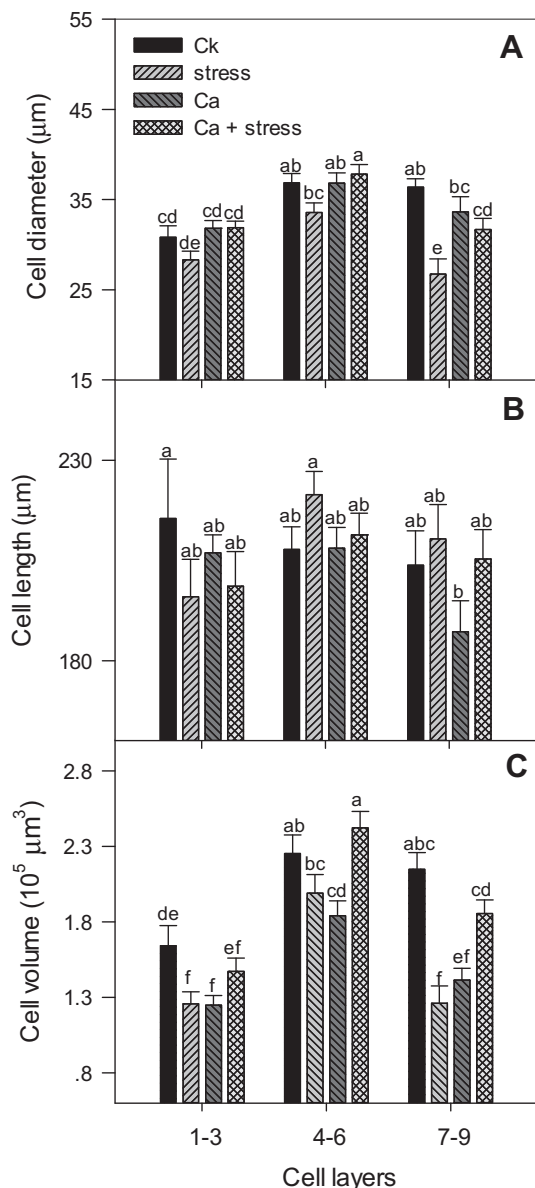


Fig. 5. Dimensions of the cortical cell layers in maize seedling primary roots. CK = 1 mM Ca^{2+} in nutrient solution; stress = 10% PEG-induced water stress in 1 mM Ca^{2+} nutrient solution; Ca = 11 mM Ca^{2+} ; Ca + stress = 10% PEG-induced water stress in 11 mM Ca^{2+} nutrient solution. A, cell diameter; B, cell length; C, cell volume. The data are the means $\pm$ SE and $N = 18-23$. The different lowercase letters above the bars denote significant differences ($P < 0.05$).

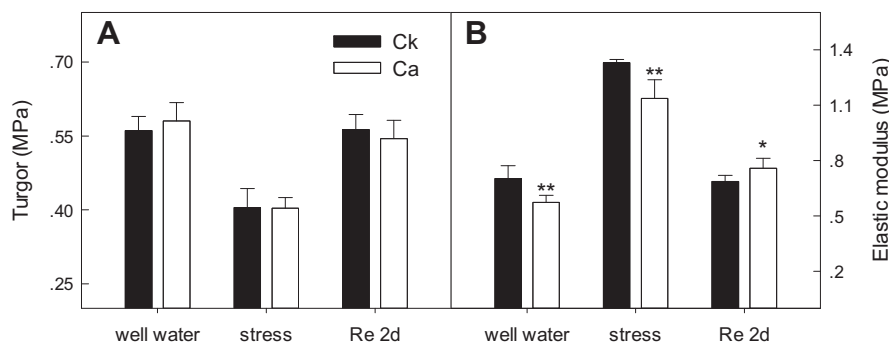


Fig. 6. Cortical cell turgor (A) and cell wall elastic modulus (B) in maize seedling primary roots. CK (black box) = 1 mM Ca^{2+} in nutrient solution and Ca (white box) = 11 mM Ca^{2+} . Stress = 10% PEG-induced water stress and Re 2d = rehydration for 2 days. The data are the means of 8 plants with SE. Levels of significance: * $P \leq 0.05$; ** $P < 0.01$.

pathway in whole hydraulic conductivity [30]. Under well-watered conditions, Hg^{2+} reduced the L_{p_r} similarly in both the control Ca^{2+} and extra Ca^{2+} roots (Fig. 3A), but it induced a larger reduction in the $L_{p_{\text{cell}}}$ of the control Ca^{2+} cortical cells (Fig. 7A). Therefore, under well-watered conditions, the primary reason for the reduced L_{p_r} in the extra Ca^{2+} plants was the decreased $L_{p_{\text{cell}}}$ of its cortical cells, which may have occurred due to the lower abundance and activity of aquaporins.

When water in xylem flows through vessel-to-vessel connections, cations (especially Ca) may reduce the swelling of pit membrane hydrogels (pectins) and subsequently decrease the xylem hydraulic resistance [31]. This “ionic effect” was reported to help to compensate for the cavitation-induced loss of hydraulic conductivity [32]. For the well-watered plants in our study, the L_{p_r} decreased significantly with the higher root Ca content that resulted from the extra Ca^{2+} treatment. Therefore, the increased Ca content may not have been located in the xylem sap, or the water resistance of the endodermis and cortex cells can dominate the L_{p_r} but not the “ionic effect”.

Similar to the salt stress results [19], PEG-induced osmotic stress contributed to a significant decline in L_{p_r} at a control Ca^{2+} concentration; however, extra Ca^{2+} (11 mM) in the nutrient solution largely reduced this decline (Fig. 3B). This result may have occurred because extra Ca^{2+} increased Ca accumulation in the apoplast or vacuoles [2], and this process may help Ca function as a cell signal. The reversible phosphorylation of aquaporins was shown to be sensitive to micro changes in the $[\text{Ca}^{2+}]_{\text{apo}}$ [15]. The contribution of the two water flow pathways may differ based on the environmental conditions [33]. In the control Ca^{2+} plants, Hg^{2+} caused a much smaller decline in L_{p_r} (53.2%) under osmotic stress conditions than (62.2%) in the well-watered conditions (Fig. 3AB). This result suggests that water deficit can decrease water flow in the apoplastic and cell-to-cell pathways, but the cell-to-cell water pathway demonstrates a relatively larger decline. Similar results obtained in $L_{p_{\text{cell}}}$ (Fig. 7AB) further confirmed this hypothesis. However, the inhibition effect of Hg^{2+} on water permeability was significantly larger in the extra Ca^{2+} plants than in the control Ca^{2+} plants during water stress (Fig. 3B; Fig. 7B). Water stress can increase the activity of Hg-sensitive water channels or up-regulate the expression levels of some aquaporins [34,35]; extra Ca^{2+} may facilitate these processes. The contribution of extra Ca^{2+} to plasma membrane stability was also refuted [20]. Thus, extra Ca^{2+} in the nutrient solution enhanced the contribution of the cell-to-cell pathway in water-stressed roots primarily due to the expression or activity levels of AQPs. The higher g_s and E values of the extra Ca^{2+} plants (Fig. 2) support this conclusion.

Osmotic stress decreased the cortical cell volumes in the control Ca^{2+} plants, whereas extra Ca^{2+} increased the cell volumes

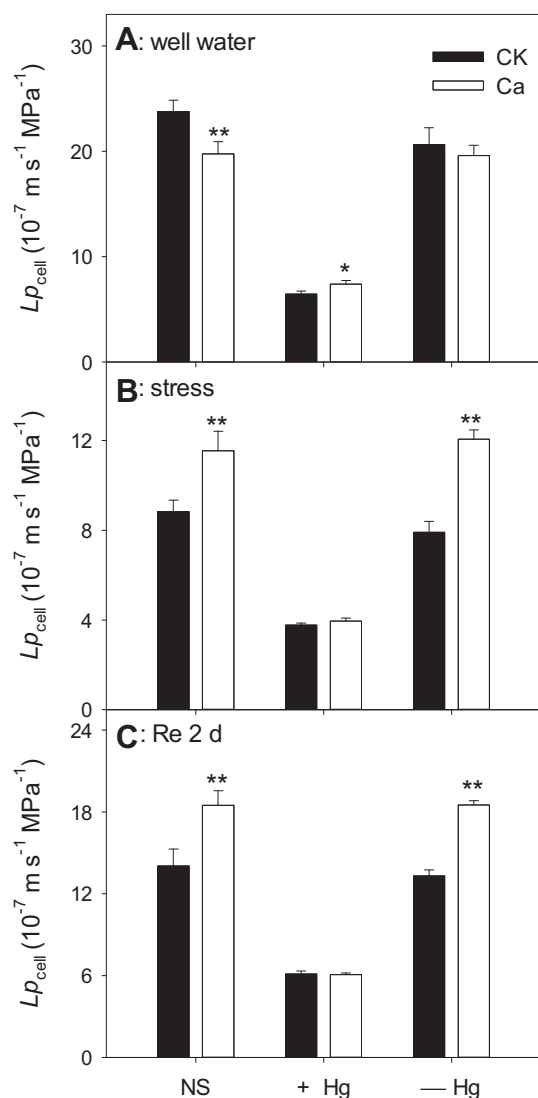


Fig. 7. Inhibition of the cortical cell hydraulic conductivity (Lp_{cell}) of maize primary roots by Hg^{2+} . A, well-watered conditions; B, 10% PEG-6000-induced water stress; C, rehydration for 2 days. CK (black box) = 1 mM Ca^{2+} in nutrient solution and Ca (white box) = 11 mM Ca^{2+} . NS = nutrient solution; +Hg = nutrient solution with 50 μM Hg^{2+} ; -Hg = Hg^{2+} was removed from the nutrient solution containing 5 mM β -mercaptoethanol. The data are the means $\pm$ SE and $N = 7$ or 8. Levels of significance: * $P \leq 0.05$; ** $P < 0.01$.

(Fig. 5C). This phenomenon may correspond to the enhanced contribution of the cell-to-cell pathway in the presence of extra Ca^{2+} during osmotic stress (Fig. 3B; Fig. 7B). Thus, a higher cell turgor and/or a lower cell wall elastic modulus were expected in extra Ca^{2+} plants, but extra Ca^{2+} did not significantly affect cell turgor (Fig. 6A). A lower cell wall elastic modulus was observed in the extra Ca^{2+} plants during osmotic stress (Fig. 6B); these plants displayed enhanced cell wall elastic ability.

Extra Ca^{2+} also helped the root hydraulic conductivity to recover during rehydration. After rehydration for 2 d, the Lp_r and Lp_{cell} partially recovered (approximately 59%) in the control plants, whereas those of the extra Ca^{2+} plants recovered almost completely (97.8% and 93.6%, respectively) (Fig. 3AC; Fig. 7AC). Lignified and/or suberized endodermal and exodermal cell walls were not different in the extra Ca^{2+} plants (Supplementary Fig. B), and morphological and anatomical changes were not detected within 2 d of rehydration (Fig. 4; Fig. 5). The Hg^{2+} experiments also

confirmed the increased contribution of the cell-to-cell pathway, which was attributed to the up-regulation of AQP expression and/or activity levels. AQPs were involved in the recovery of hydraulic conductance during rehydration [13,36], and the role of plasma membrane intrinsic proteins (PIPs) in the xylem parenchyma was emphasized [37]. In the control Ca^{2+} plants, rehydration increased the contribution of the cell-to-cell pathways because Hg^{2+} similarly inhibited the increased Lp_r values compared with stressed plants. In the extra Ca^{2+} plants, the Hg^{2+} inhibition ratio of the Lp_r was lower during rehydration (62.4%) than during osmotic stress (74.6%). Although Hg^{2+} inhibited the Lp_r more seriously in the extra Ca^{2+} plants ($3.1 \times 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$) than in the control Ca^{2+} plants ($2.0 \times 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$), extra Ca^{2+} reduced the relative contribution of the cell-to-cell pathway during rehydration.

4. Conclusion

Extra Ca^{2+} in the nutrient solution may regulate the relative contribution of water flow pathways in maize roots. Under well-watered conditions, extra Ca^{2+} increased the root Ca content and significantly decreased the hydraulic conductivity of the whole root and cortical cells. Osmotic stress decreased leaf water loss, Lp_r , and Lp_{cell} significantly in the control Ca^{2+} plants; however, extra Ca^{2+} weakened these negative responses. The data from the Hg^{2+} inhibition experiments indicated a larger contribution of the cell-to-cell pathway in the extra Ca^{2+} plant roots during osmotic stress due to AQP activity or expression levels. Moreover, extra Ca^{2+} helped the leaf gas exchange and root hydraulic conductivity to recover during rehydration. Overall, extra Ca^{2+} may regulate the activity and/or expression levels of AQPs to increase the contribution of the cell-to-cell pathway during osmotic stress. However, under well-watered or rehydration conditions, the contribution of the cell-to-cell pathway decreased in the extra Ca^{2+} plants. And extra Ca^{2+} plants can show a relative stable water status during variable water conditions.

5. Materials and methods

5.1. Plant material and growth conditions

Maize seeds (*Zea mays* L., cv. HuDan 4) were sterilized in 5% NaClO for 10 min; they were then thoroughly rinsed and germinated on wet filter paper in the dark at 29 °C. Two days after sowing, the seedlings were transferred into 6-L plastic cylindrical containers (approximately six seedlings per container) containing well-aerated, half-strength modified Hoagland nutrient solution (pH 5.5) [38]. The plants were grown in an artificial climate chamber at 27/22 °C, 12 h light, 60% RH, and photosynthetic active radiation (PAR) of 500 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$. Half of the plants were grown in control Ca concentrations (1 mM), and the other half were grown in the presence of extra Ca (11 mM). Three days after the transfer, osmotic stress (-0.3 MPa) was induced by adding 10% polyethylene glycol 6000 (PEG-6000); the stress was maintained for approximately 7 d until the plants reached the three-leaf stage, and they were then rehydrated for 2 d. Measurements were conducted at the end of the water stress and rehydration treatments.

5.2. Calcium analysis

After cutting the lateral roots, the primary radicle roots were divided into the three following portions: 0–1 cm, 1–10 cm, and 10–35 cm from the root apex. The Ca content was analyzed by atomic absorption spectrophotometry (AAS) [39]. Briefly, samples were dried under 110 °C for 0.5 h and then at 80 °C for approximately 48 h. Approximately 1.0 g of crushed powder of each sample

was dry-ashed in a crucible. Approximately 20 mL of 1.2 M hydrochloric acid was added to dissolve the ash before the mixture was filtered. Next, 1.0 mL of the filtrate and 1.0 mL of LaCl_3 were diluted and combined into a 50 mL volume, and the Ca content was determined using an atomic absorption spectrophotometer (ZL5100, PerkinElmer Inc., USA). A series of standard CaCl_2 solutions was used to quantify the Ca content. Three independent replications were taken for each measurement.

5.3. Leaf gas exchange

The leaf transpiration rate (E) and stomatal conductance (g_s) were measured on the third leaf at noon using a portable gas-exchange system (LI-6400, LI-CRO Inc., Lincoln, NE, USA). Five or six random plants were measured per treatment.

5.4. Morphology and anatomy

The root surface area, total root length, and total root surface area were determined using a scanner and WinRHIZO PRO 2009 (Regent Inc., Canada). The primary root diameter was measured at a location 2 cm from the root apex. Freehand cross- and longitudinal- sections at distances of 3–5 cm from the primary root apex were made to determine the mean dimensions of the cortical cells. The sections were stained with 0.5% Toluidine Blue O for 3 min and rinsed several times in distilled water before mounting; images were taken and measured with the Image-Pro-Express 6.0 software program. For each cortical root layer, the mean area and volume of the cortical cells were determined from the diameters and lengths of the cells.

5.5. Whole root hydraulic conductivity (L_p)

A pressure chamber was used to measure the hydraulic conductance (L_p) of the whole root system. The L_p measurements were performed at noon. Maize roots were detached at the upper end of the mesocotyl and introduced into the cylindrical pressure chamber that contained half-strength Hoagland solution. When the water-stressed roots were measured, PEG was rinsed carefully with the same nutrient solution. The method was based on the differences in sap flow volume exuded over a given time interval at different external root pressures. The root xylem sap that was exuded at pressures of 0.1, 0.2, 0.3 and 0.4 MPa was collected in Eppendorf tubes for 60 s (t), and the volume (V) was measured. The flow rate, J_v , was also determined as follows:

$$J_v = V/(A_r/t) \quad (1)$$

After the measurements were obtained, the root surface area (A_r) was determined using a scanner and WinRHIZO PRO 2009 (Regent Inc., Canada). The slope of the xylem flow (J_v) that was calculated using Eq. (1) was plotted against the applied pressure to provide the L_p value as follows:

$$L = J_v/\Delta P \quad (2)$$

5.6. Cortical cell hydraulic conductivity

The hydraulic conductivity of the cortical cells ($L_{p_{\text{cell}}}$) was measured using a cell pressure probe [40]. The measurements were made on the fourth to sixth layer of cortical cells at the same anatomical site. To determine the activity levels of the aquaporins during water flow across the roots, 50 μM HgCl_2 was added to the nutrient solution after measuring L_p or $L_{p_{\text{cell}}}$. After 20 min of Hg^{2+} treatment, fresh, half-strength Hoagland nutrient solution was

supplied, and L_p or $L_{p_{\text{cell}}}$ was measured again. After removing Hg^{2+} from roots with β -mercaptoethanol (5 mM; 10 min), the L_p or $L_{p_{\text{cell}}}$ was measured again.

5.7. Data analysis

The data were analyzed using a one-way ANOVA, and the treatment means were compared using independent-sample t -tests.

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Appendix A. Supplementary material

Supplementary material associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.plaphy.2012.07.005>.

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